5.2: Carbohydrates serve as fuel and building material

Carbohydrates include sugars and polymers of sugars. The simplest carbohydrates are the monosaccharides, or simple sugars; these are the monomers from which more complex carbohydrates are built. Disaccharides are double sugars, consisting of two monosaccharides joined by a covalent bond. Carbohydrate macromolecules are polymers called polysaccharides, composed of many sugar building blocks.

Animation

Carbohydrates

image

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Sugars

Monosaccharides (from the Greek *monos*, single, and *sacchar*, sugar) generally have molecular formulas that are some multiple of the unit CH_2O. Glucose $C_6H_{12}O_6$, the most common monosaccharide, is of central importance in the chemistry of life. In the structure of glucose, we can see the trademarks of a monosaccharide: The molecule has a carbonyl group, ╲╱C═O, and multiple hydroxyl groups, —OH** (**Figure 5.3**)**. Depending on the location of the carbonyl group, a monosaccharide is either an aldose (aldehyde sugar) or a ketose (ketone sugar). Glucose, for example, is an aldose; fructose, an isomer of glucose, is a ketose. (Most names for sugars end in -*ose*.) Another criterion for classifying monosaccharides is the size of the carbon skeleton, which ranges from three to seven carbons long. Glucose, fructose, and other sugars that have six carbons are called hexoses. Trioses (three-carbon sugars) and pentoses (five-carbon sugars) are also common.

Figure 5.3

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The structure and classification of some monosaccharides.
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Structure and classification of some monosaccharides.

Sugars vary in the location of their carbonyl groups (orange), the length of their carbon skeletons, and the way their parts are arranged spatially around asymmetric carbons (compare, for example, the purple portions of glucose and galactose).

Sugars can be classified by the number of carbons, the location of the carbonyl group, and the spatial arrangement at each asymmetric carbon.

The number of carbons in a sugar molecule is one way to classify it. For example, a triose is a sugar with three carbons and the formula C 3 H 6 O 3. Examples include glyceraldehyde and dihydroxyacetone, which are both initial breakdown products of glucose. A pentose has five carbons and the formula C 5 H 10 O 5. Examples of pentoses are ribose, a component in RNA and ribulose, an intermediate in photosynthesis. A hexose has six carbons and the formula C 6 H 12 O 6. Glucose, galactose and fructose are hexoses used as energy sources for organisms.

The location of the carbonyl group (C double bonded to O) on the carbon chain is another way to further classify a sugar. A sugar that has its carbonyl group at the end of the chain (an aldehyde) is called an aldose. Glyceraldehyde, ribose, glucose and galactose are aldoses. When the carbonyl group is not on the end of the carbon chain, it is a ketone and the sugar is called a ketose. Dihydroxyacetone, ribulose and fructose are ketoses.

The position of the hydroxyl groups (O H) on each asymmetric carbon is the most specific way to classify sugars. For example, glucose and galactose both have six carbons. Drawn vertically with a linear carbon skeleton (Fischer projection), the first (top) carbon is part of an aldehyde (C H O). The remaining carbons have O H groups on either the left or the right side of the chain, and a hydrogen atom across from each O H. The only difference between the two sugars is the location of the O H on the fourth carbon. This O H is on the right in glucose, but on the left in galactose. In all other ways, both sugars are the same. The second and fifth carbons have an OH on the right. The third carbon has O H on the left. The sixth carbon is not asymmetric, and has two hydrogens and an O H. Fructose has the same structure as glucose, but, since it is a ketose, it has only a carbonyl group on carbon 2. The first carbon in fructose is a C H 2 O H, which is not asymmetric.

MAKE CONNECTIONS *In the 1970s, a process was developed that converts the glucose in corn syrup to its sweeter-tasting isomer, fructose. High-fructose corn syrup, a common ingredient in soft drinks and processed food, is a mixture of glucose and fructose. What type of isomers are glucose and fructose? (See*Figure 4.7*.)*

*For suggested answer, see*Appendix A*.*

Animation

Monosaccharides

image

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Still another source of diversity for simple sugars is in the way their parts are arranged spatially around asymmetric carbons. (Recall that an asymmetric carbon is a carbon attached to four different atoms or groups of atoms.) Glucose and galactose, for example, differ only in the placement of parts around one asymmetric carbon (see the purple boxes in **Figure 5.3**). What seems like a small difference is significant enough to give the two sugars distinctive shapes and binding activities, thus different behaviors.

Although it is convenient to draw glucose with a linear carbon skeleton, this representation is not completely accurate. In aqueous solutions, glucose molecules, as well as most other five- and six-carbon sugars, form rings, because they are the most stable form of these sugars under physiological conditions **(Figure 5.4)**.

Figure 5.4

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Linear and ring forms of glucose.
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The illustration gives the linear and ring forms of glucose.

Chemical equilibrium between the linear and ring structures greatly favors the formation of rings. The carbons of the sugar are numbered 1 to 6. To form the glucose ring, carbon 1 bonds to the oxygen attached to carbon 5. The linear form is a vertical chain where carbons are numbered from top to bottom. C 1 is double bonded to O up and right, and single bonded to H up and left. Carbons 2 through 6 are single bonded to H and single bonded to O H, and C 6 is also single bonded to another H. When a ring forms, the O from O H on C 5 bonds to C 1. The H from that O H then moves to the double bonded O from C 1. This forms a single bond to O H from C 1. The ring has 5 C atoms and 1 O atom. C 5 is single bonded to C H 2 O H, where that C is C 6.

DRAW IT *Start with the linear form of fructose (see*Figure 5.3*)* and draw the formation of the fructose ring in two steps, as shown in (a). First, number the carbons starting at the top of the linear structure. Then draw the molecule in a ringlike orientation, attaching carbon*5*via its oxygen to carbon*2*. Compare the number of carbons in the ring portions of fructose and glucose.*

*For suggested answer, see**Appendix A*.

Monosaccharides, particularly glucose, are major nutrients for cells. In the process known as cellular respiration, cells extract energy from glucose molecules by breaking them down in a series of reactions. Not only are monosaccharides a major fuel for cellular work, but their carbon skeletons also serve as raw material for the synthesis of other types of small organic molecules, such as amino acids and fatty acids. Monosaccharides that are not immediately used in these ways are generally incorporated as monomers into disaccharides or polysaccharides, discussed next.

A **disaccharide** consists of two monosaccharides joined by a **glycosidic linkage**, a covalent bond formed between two monosaccharides by a dehydration reaction (*glyco* refers to carbohydrate). For example, maltose is a disaccharide formed by the linking of two molecules of glucose **(Figure 5.5a)**. Also known as malt sugar, maltose is an ingredient used in brewing beer. The most prevalent disaccharide is sucrose, or table sugar. Its two monomers are glucose and fructose **(Figure 5.5b)**. Plants generally transport carbohydrates from leaves to roots and other nonphotosynthetic organs in the form of sucrose. Lactose, the sugar present in milk, is another disaccharide, in this case a glucose molecule joined to a galactose molecule. Disaccharides must be broken down into monosaccharides to be used for energy by organisms. Lactose intolerance is a common condition in humans who lack lactase, the enzyme that breaks down lactose. The sugar is instead broken down by intestinal bacteria, causing formation of gas and subsequent cramping. The problem may be avoided by taking the enzyme lactase when eating or drinking dairy products or consuming dairy products that have already been treated with lactase to break down the lactose.

Figure 5.5

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Examples of disaccharide synthesis.
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Abbreviated ring structure and Dehydration reaction in the synthesis of (a) maltose and (b)sucrose.

Abbreviated ring structure. Each unlabeled corner represents a carbon. The ring's thicker edge indicates that you are looking at the ring edge-on; the components attached to the ring lie above or below the plane of the ring. The ring form of glucose is drawn using an edge-on view of a hexagon. The points of the hexagon are on the right and left, and the carbons are numbered starting with the carbon on the far right, which was the aldehyde carbon in the linear form of the sugar. Each carbon is bonded to an O H and a hydrogen, one above the ring, and the other below it. Carbons 2 and 3 are on the near side of the hexagon, carbon 4 is the left point, and carbon 5 is on the far side. The oxygen of carbon 5 completes the hexagon, as it is bonded to carbon 1. Carbon 6, which is a C H 2 O H is bonded to carbon 5 and drawn above the ring. In glucose, the O H groups are below the ring on carbons 2 and 4, and above the ring on carbon 3. The O H on carbon 1 can be

either below or above the ring, depending on whether the ring is in the alpha or beta form, respectively (see Figure 5.7a).

(a)Dehydration reaction in the synthesis of maltose. The bonding of two glucose units forms maltose. The 1-4 glycosidic linkage joins the number 1 carbon of one glucose to the number 4 carbon of the second glucose. Joining the glucose monomers in a different way would result in a different disaccharide. Two glucose molecules can form a bond to make a disaccharide. In the case of maltose, a 1-4 glycosidic linkage is formed between two glucose molecules. This means that carbon 1 of the first glucose molecule is bonded to the oxygen on carbon 4 for the second glucose molecule. When this bond forms, the O H on carbon 1 of the first glucose and the H from the hydroxyl group on the second glucose leave and make a water molecule. This dehydration reaction leads to the formation of a disaccharide. If two glucose molecules bonded in a different pattern, a different disaccharide would result.

(b)Dehydration reaction in the synthesis of sucrose. Sucrose is a disaccharide formed from glucose and fructose. Notice that fructose forms a five-sided ring, though it is a hexose-like glucose. The structure of fructose is as follows: Each unlabeled corner represents a carbon. The ring's thicker edge indicates that you are looking at the ring edge-on; the components attached to the ring lie above or below the plane of the ring. The ring form of fructose is drawn using an edge-on view of a pentagon. The points of the pentagon are on the right and left, and the carbons are numbered starting with the carbon on the far right. Each carbon is bonded to an O H and hydrogen, one above the ring, and the other below it. Carbons 3 and 4 are on the near side of the hexagon, carbon 5 is the left point, and carbon 6 is on the far side. Carbon 1 is part of C H 2 O H bonded to carbon 2 of the ring. The oxygen of carbon 5 completes the pentagon, as it is bonded to carbon 1. Carbon 6, which is a C H 2 O H is bonded to carbon 5 and drawn above the ring. In fructose, the O H groups are below the ring on carbons 3 and 4. The bonding of one glucose unit and one fructose unit forms sucrose. The 1-2 glycosidic linkage joins the number 1 carbon of glucose to the number 2 carbon of fructose. Joining the monomers in a different way would result in a different disaccharide. When this bond forms, the O H on carbon 1 of the glucose unit and the H from the hydroxyl group on the fructose unit leave and make a water molecule. This dehydration reaction leads to the formation of a disaccharide.

DRAW IT *Referring to**Figures 5.3**and*5.4*, number the carbons in each sugar in this figure. How does the name of each linkage relate to the numbers?*

*For suggested answer, see**Appendix A*.

Animation

Synthesis of Sucrose

image

Play

image

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Polysaccharides

Polysaccharides are macromolecules, polymers with a few hundred to a few thousand monosaccharides joined by glycosidic linkages. Some polysaccharides serve as storage material, hydrolyzed as needed to provide monosaccharides for cells. Other polysaccharides serve as building material for structures that protect the cell or the whole organism. The architecture and function of a polysaccharide are determined by its monosaccharides and by the positions of its glycosidic linkages.

Storage Polysaccharides

Both plants and animals store sugars for later use in the form of storage polysaccharides **(Figure 5.6)**. Plants store **starch**, a polymer of glucose monomers, as granules within cellular structures known as plastids. (Plastids include chloroplasts.) Synthesizing starch enables the plant to stockpile surplus glucose. Because glucose is a major cellular fuel, starch represents stored energy. The sugar can later be withdrawn by the plant from this carbohydrate “bank” by hydrolysis, which breaks the bonds between the glucose monomers. Most animals, including humans, also have enzymes that can hydrolyze plant starch, making glucose available as a nutrient for cells. Potato tubers and grains—the fruits of wheat, maize (corn), rice, and other grasses—are the major sources of starch in the human diet.

Figure 5.6

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Polysaccharides of plants and animals.
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Gardener holding potatoes; arrows flow to microscopic image depicting storage structures called plastids containing starch granules in potato tuber cell. Diagram of starch structure flows from image.

Two starch structures depicted: amylose starch and amylopectin.

Amylose starch consists of a long, unbranched chain of glucose (6 carbon ring) monomers joined by 1-4 linkages (number 1 carbon to number 4 carbon).

Amylopectin starch consists of a somewhat branched chain of glucose monomers with 1-6 linkages at the branch points. One branch is depicted in the diagram.

Arrow flows from gardener's arm to microscopic image of glycogen granules stored in muscle tissues (scale: 50 micrometers). Diagram of glycogen structure flows from image. Glycogen is a branched polysaccharide composed of glucose monomers. Several branches depicted in the diagram, indicating that glycogen is more extensively branched than amylopectin starch. Arrow flows from leaf to microscopic image of plant cell with cell wall visible; another arrow flows from cell wall to image of cellulose. Structure of cellulose flows from image.

(a) Starch stored in plant cells, **(b)** glycogen stored in muscle cells, and **(c)** structural cellulose fibers in plant cell walls are all polysaccharides composed entirely of glucose monomers (green hexagons). In starch and glycogen, the polymer chains tend to form helices in unbranched regions because of the angle of the 1–4 linkages between glucose molecules. There are two kinds of starch: amylose and amylopectin. Cellulose, with a different kind of glucose linkage, is always unbranched.

Most of the glucose monomers in starch are joined by 1–4 linkages (number 1 carbon to number 4 carbon), like the glucose units in maltose (see Figure 5.5a). The simplest form of starch, amylose, is unbranched. Amylopectin, a more complex starch, is a branched polymer with 1–6 linkages at the branch points. Both of these starches are shown in **Figure 5.6a**.

Animals store a polysaccharide called **glycogen**, a polymer of glucose that is like amylopectin but more extensively branched **(Figure 5.6b)**. Vertebrates store glycogen mainly in liver and muscle cells. Breakdown of glycogen in these cells releases glucose when the demand for energy increases. (The extensively branched structure of glycogen fits its function: More free ends are available for breakdown.) This stored fuel cannot sustain an animal for long, however. In humans, for example, glycogen stores are depleted in about a day unless they are replenished by eating. This is an issue of concern in low-carbohydrate diets, which can result in weakness and fatigue.

Structural Polysaccharides

Organisms build strong materials from structural polysaccharides. For example, the polysaccharide called **cellulose** is a major component of the tough walls that enclose plant cells **(Figure 5.6c)**. Globally, plants produce almost 1014 kg 100 billion tons of cellulose per year; it is the most abundant organic compound on Earth.

Like starch, cellulose is a polymer of glucose with 1–4 glycosidic linkages, but the linkages in these two polymers differ. The difference is based on the fact that there are actually two slightly different ring structures for glucose **(Figure 5.7a)**. When glucose forms a ring, the hydroxyl group attached to the number 1 carbon is positioned either below or above the plane of the ring. These two ring forms for glucose are called alpha α and beta β, respectively. (Greek letters are often used as a "numbering" system for different versions of biological structures, much as we use the letters a, b, c, and so on for the parts of a

question or a figure.) In starch, all the glucose monomers are in the α configuration **(Figure 5.7b)**, the arrangement we saw in Figures 5.4 and 5.5. In contrast, the glucose monomers of cellulose are all in the β configuration, making every glucose monomer "upside down" with respect to its neighbors (**Figure 5.7c**; see also Figure 5.6c).

Figure 5.7

Starch and cellulose structures.

Starch and cellulose structures.

Part a. Alpha and beta glucose ring structures. These two interconvertible forms of glucose differ in the placement of the hydroxyl group attached to the number 1 carbon. Whenever a sugar forms a ring in solution, there are two possible structures, depending on how the ring closes. For example, in glucose, if the ring closes with the OH on carbon 1 oriented down, then the sugar is in the alpha form. If the ring closes with the OH on this carbon pointed up, then the sugar is in the beta form. The two forms can open up and close again in solution, so they are in chemical equilibrium with each other and the linear form of the sugar.

Part b. Starch: 1-4 linkage of alpha glucose monomers. All monomers are in the same orientation. Starch is a polysaccharide of alpha glucose. Starch is the result of assembling alpha glucose into a polysaccharide using 1-4 glycosidic bonds. All monomers are in the same orientation. The resulting starch molecule has an overall helical shape.

Part c. Cellulose is a polysaccharide of beta glucose with 1-4 glycosidic linkages. Every beta glucose monomer is upside down with respect to its neighbors. Cellulose is the result of assembling beta glucose into a polysaccharide using 1-4 glycosidic bonds. Every beta glucose monomer is upside down with respect to its neighbors. The resulting cellulose molecule is straight and rather rigid.

Animation

Starch, Cellulose, and Glycogen Structures

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The differing glycosidic linkages in starch and cellulose give the two molecules distinct three-dimensional shapes. Certain starch molecules are largely helical, fitting their function of efficiently storing glucose units. Conversely, a cellulose molecule is straight. Cellulose is never branched, and some hydroxyl groups on its

glucose monomers are free to hydrogen-bond with the hydroxyls of other cellulose molecules lying parallel to it. In plant cell walls, parallel cellulose molecules held together in this way are grouped into units called microfibrils (see Figure 5.6c). These cable-like microfibrils are a strong building material for plants and an important substance for humans because cellulose is the major constituent of paper and the only component of cotton. The unbranched structure of cellulose thus fits its function: imparting strength to parts of the plant.

Enzymes that digest starch by hydrolyzing its α linkages are unable to hydrolyze the β linkages of cellulose due to the different shapes of these two molecules. In fact, few organisms possess enzymes that can digest cellulose. Almost all animals, including humans, do not; the cellulose in our food passes through the digestive tract and is eliminated with the feces. Along the way, the cellulose abrades the wall of the digestive tract and stimulates the lining to secrete mucus, which aids in the smooth passage of food through the tract. Thus, although cellulose is not a nutrient for humans, it is an important part of a healthy diet. Most fruits, vegetables, and whole grains are rich in cellulose. On food packages, “insoluble fiber” refers mainly to cellulose (see Figure 5.7).

Some microorganisms can digest cellulose, breaking it down into glucose monomers. A cow harbors cellulose-digesting prokaryotes and protists in its gut. These microbes hydrolyze the cellulose of hay and grass and convert the glucose to other compounds that nourish the cow. Similarly, a termite, which is unable to digest cellulose by itself, has prokaryotes or protists living in its gut that can make a meal of wood. Some fungi can also digest cellulose in soil and elsewhere, thereby helping recycle chemical elements within Earth’s ecosystems.

Another important structural polysaccharide is **chitin**, the carbohydrate used by arthropods (insects, spiders, crustaceans, and related animals) to build their exoskeletons—hard cases that surround the soft parts of an animal **(Figure 5.8)**. Made up of chitin embedded in a layer of proteins, the case is leathery and flexible at first, but becomes hardened when the proteins are chemically linked to each other (as in insects) or encrusted with calcium carbonate (as in crabs). Chitin is also found in fungi, which use this polysaccharide rather than cellulose as the building material for their cell walls. Chitin is similar to cellulose, with β linkages, except that the glucose monomer of chitin has a nitrogen-containing attachment (see Figure 5.8).

Figure 5.8

Chitin, a structural polysaccharide.

Chitin, a structural polysaccharide. Part a. The chitin monomer structure. Part b. Chitin in nature.

Part a. The chitin monomer is N-acteylglucosamine, a derivative of glucose where the alcohol on carbon 2 is replaced with an acetamido group.

Part b. Chitin, embedded in proteins, forms the exoskeleton of arthropods. This picture shows a cicada molting, shedding its old exoskeleton and emerging in adult form.